

Locally Biased, Financially Blind: Where State Investor Literacy Moderates the Momentum–Idiosyncratic-Volatility Anomaly

Adam Bozman*

May 19, 2026

Abstract

Retail investors trade locally, and they trade more like the people around them than the textbook investor: chasing recent winners, anchoring on familiar names, paying limited attention to volatility-adjusted risk. State-level financial literacy varies substantially across the United States, and a large household-finance literature documents that less literate investors make systematically worse portfolio choices. If those choices show up in prices, they should show up as a state-literacy moderation of return anomalies that depend on retail mispricing. Yet in aggregate, the empirical signature is faint: a regression

*Washington State University. Email: bozmanadam@gmail.com.

of the momentum–idiosyncratic-volatility interaction on state literacy returns modest coefficients without a clean economic story. This paper asks a sharper question: not whether literacy moderates the anomaly, but *where in the cross-section* it does, and what microfoundation that location identifies. The answer, on the stable-headquarters subsample of CRSP-listed U.S. common stock from 2009 to 2023, is that the moderation lives in firms that occupy an interaction zone — enough retail demand to move prices, enough institutional activity to translate that demand into return differences across literate and illiterate states. In the full panel these firms fall in the middle of the institutional-ownership distribution; under sample restrictions to larger firms they migrate to the low-IO tercile, exactly as the interaction-zone reading predicts. Firms dominated by institutions show no literacy gradient at all, consistent with arbitrage neutralizing the local-state channel. The contribution is a precisely-bounded characterization of where a literacy mechanism operates, and an empirical signature — migration with size composition — that identifies the mechanism without requiring an instrumental-variables strategy.

1 Introduction

Lusardi and Mitchell [2014] and a large literature that follows them document that financially literate households make systematically better portfolio decisions: they diversify more, panic less, time the market less, and

pay more attention to fees and volatility-adjusted risk. The variation across U.S. states is substantial. In high-literacy states like Minnesota, Hawaii, and New Hampshire, roughly 28 percent of adults can answer three of five basic literacy questions correctly; in Mississippi, Ohio, and West Virginia, the share is around 21 percent. If the trading decisions of less literate investors leave a footprint in equilibrium prices, that footprint should be visible as a state-level moderator of the return anomalies most plausibly driven by retail behavior — momentum, idiosyncratic volatility, and their interaction.

The empirical literature on local bias provides the second piece of the setup. Coval and Moskowitz [1999] and Ivković and Weisbenner [2005] document that retail investors hold portfolios disproportionately weighted toward locally headquartered firms; Hong et al. [2008] show that this local demand moves prices for the firms in question; Bernile et al. [2015] extend the result to other dimensions of geographic clientele. The natural prediction follows: where retail investors are less financially literate, they should trade more like the textbook noise trader, and the anomalies their behavior generates should be stronger. State NFCS literacy, applied to firms by headquarters state, gives the moderator; the momentum-IV anomaly interaction, which combines a retail-friendly anomaly with the standard limits-to-arbitrage measure, gives the natural object to study.

Putting these pieces together produces a clean hypothesis: state literacy should moderate the momentum-IV interaction, with the effect weaker in literate states and stronger in illiterate states. In aggregate, however, the

data do not deliver. Regressing the three-way interaction on the full panel of CRSP-listed common stock returns a modest negative coefficient with marginal statistical significance, and the magnitude collapses by roughly forty fold once headquarters-state reassignment is corrected for the 14 percent of firm-months whose static Compustat snapshot does not match their then-current 10-K filing. The aggregate test is misleading in two ways. It pools firms with very different ownership structures, and it pools firms whose local-investor base is genuinely defined (firms with stable headquarters across the panel) with firms whose local-investor base is in transition (relocators, for whom neither origin nor destination state correctly characterizes the marginal investor). The aggregate null is therefore not informative about whether a literacy channel exists. It is informative about whether the channel operates uniformly across the cross-section. It does not.

The contribution of this paper is to ask a sharper question. Where in the cross-section does the literacy mechanism become visible? Under what conditions on firm ownership and size does retail mispricing translate into price differences that can be detected across literate and illiterate states? And what does the answer identify about the microfoundation?

The data deliver a sharp pattern. On the stable-headquarters subsample of CRSP-listed common stock from 2009 to 2023, the literacy moderation of the momentum-IV interaction concentrates in firms whose institutional ownership sits in the middle of the within-stock IO distribution — the mid-IO tercile, with mean ownership share around fifty percent. Firms with predom-

inantly retail ownership (low IO) show essentially no literacy moderation. Firms with predominantly institutional ownership (high IO) also show none. The mid-tercile coefficient survives the addition of firm-size moderation as a control, so the finding is not a size effect dressed in ownership clothing, even though mid-IO firms are predominantly mid-cap. Within mid-IO, the coefficient corresponds to a cross-state literacy gradient that recovers exactly the prediction the hypothesis makes: states with higher NFCS literacy show weaker anomaly slopes, and the across-state correlation between per-state coefficients and state literacy is sharp and statistically clean.

The economic interpretation that the data identify is an interaction zone. The literacy channel requires two ingredients to leave a price footprint: retail demand large enough to move prices, and institutional activity sufficient to translate that demand into return differences detectable across states. Where neither ingredient is present (micro-cap firms held mostly by retail with no institutional counterparty), retail mispricing is genuine but does not load on cross-state literacy variation; it accumulates uniformly without an arbitrage counterpressure. Where institutions dominate (large firms with high IO), institutional arbitrage absorbs whatever retail demand exists and neutralizes the local-state channel; literacy varies, but the marginal investor doesn't. The interaction zone — the population of firms with mixed ownership at meaningful market size — is where both ingredients are present, and is where the data find the literacy moderation. In the full stable-HQ panel, the interaction zone falls predominantly in mid-IO. Under sample restrictions

to larger firms, it relocates. Specifically: dropping the bottom 50 percent of firms by market capitalization shifts the moderation from mid-IO to low-IO, and the cross-state literacy gradient migrates with it. The same economic population, relocated to a different tercile of the conditional IO distribution.

The migration is the central piece of mechanism evidence the paper introduces. It is the empirical signature an interaction-zone reading predicts and the IO-tercile result alone does not deliver. The high-IO tercile shows no cross-state literacy gradient under any sample restriction, providing a clean predicted-null check the reading passes. The IV-decile decomposition within mid-IO, which would test the canonical limits-to-arbitrage microfoundation under which the effect strengthens at high IV, fails to reject decile equality and returns a near-zero coefficient at the highest IV decile. The textbook limits-to-arbitrage reading is rejected; the interaction-zone reading, identified through size and ownership conditioning rather than through IV, is identified positively by the migration.

Three bounds discipline the result. First, the continuous-margin quadratic-IO five-percent statistical clearance for the U-shape across IO terciles does not survive log, kink, or twenty-bin alternatives; the functional-form-free claim is the discrete-tercile contrast. Second, on the corrected full panel under point-in-time literacy reassignment, the mid-IO coefficient magnitude collapses by roughly twenty-six fold; the characterization is therefore a statement about firms with stable headquarters across the panel window, not about the corrected full panel. Third, an instrumental-variables strategy

using state K-12 personal-finance graduation requirements as a source of exogenous variation in adult literacy fails its first-stage relevance test in the right direction — mandate states are systematically less literate, reflecting that states adopted mandates as a response to low adult literacy. The OLS estimate is the preferred inferential statement, and the IV failure is documented honestly.

The closest prior work is Korniotis and Kumar [2013], who show that state-level business cycles predict returns of locally headquartered firms. This paper differs on three margins: the state variable is NFCS literacy rather than the business cycle, the object of study is a three-way moderation of an anomaly interaction rather than a level prediction, and the discriminating margin is a within-stock IO contrast that the business-cycle channel does not predict. A business-cycle control block attenuates the literacy three-way by 20–23 percent, so the two state moderators are distinct but partly overlapping. Korniotis et al. [2020] use brokerage-demographic proxies for state-level sophistication, but neither NFCS literacy nor the momentum-IV interaction. Hillert et al. [2014] is the closest precedent for a state-level moderator of momentum, using media coverage as the primary moderator and state investor individualism as a cross-sectional moderator of the media-momentum effect, rather than literacy.

Roadmap. Section ?? describes the data and sample, including the construction of the NFCS state-literacy panel and the stable-headquarters sub-

sample. Section 3 states the specification and the identifying assumptions. Section 4 reports the discrete-tercile mid-IO concentration, the size-controlled extension, the migration, and the cross-state literacy gradient. Section 5 develops the interaction-zone characterization. Section 6 reports the disciplining battery: the point-in-time correction, the IV-decile decomposition, the pre/post-2015 split, the failed instrumental-variables strategy, and the functional-form grid. Section 7 discusses interpretation and limitations. Section 8 concludes.

2 Data and Sample Construction

2.1 Sample

The sample is the cross-section of CRSP-listed common stock (share codes 10 and 11; exchange codes 1, 2, 3, covering NYSE, NYSE American, and NASDAQ) from January 2009 through December 2023. The full panel contains 623,896 firm-month observations, 7,447 unique firms, 51 headquarters states (the fifty states plus the District of Columbia), and 180 calendar months. I z-score all focal variables within each calendar month, so the residual variation the regressions exploit is cross-sectional and free of time-aggregate movements. State financial literacy is measured from the NFCS Big Five literacy questions, fielded by the FINRA Investor Education Foundation in waves 2009, 2012, 2015, 2018, and 2021, and forward-filled to monthly frequency between waves.

2.2 The stable-headquarters subsample

The local-bias channel transmits through community ties between a firm and its headquarters-state retail investor base. Such ties take time to form and to decay, so for firms that relocate headquarters within sample the effective local-investor population is ambiguous by construction. I construct a stable-headquarters subsample that drops firms with a confirmed in-sample EDGAR-based relocation, using a union of two flags: `hdr_changed` fires when the 10-K-header business-address `STATE` changes across a firm’s in-sample 10-Ks (954 firms); `sec_disagree` fires when the firm’s most-recent 10-K-header state disagrees with the static Compustat snapshot and the firm has at least two parsed 10-K headers (524 additional firms). The union identifies 1,366 relocating firms (18.9 percent of flag-covered firms, 16.8 percent of firm-months). The top first-to-last state pairs are consistent with the headquarters-relocation mechanism of Pirinsky and Wang [2006], who show that relocating firms shift their local return comovement from origin to destination state. The stable-headquarters subsample contains 519,090 firm-months and 6,081 firms.

The paper’s headline patterns are estimated on the stable-headquarters subsample under static-snapshot literacy; the point-in-time literacy reassignment on the full panel is the correctness check (Section ??). Construction detail and PIT-coverage diagnostics are in IA ?? and IA ??.

2.3 Variable definitions

The dependent variable is the monthly stock return `ret` from CRSP. Momentum is `mom_12_2`, the standard Jegadeesh and Titman [1993] cumulative return from twelve months before to two months before the observation month, skipping the most recent month. Idiosyncratic volatility (IV) is the standard deviation of daily Fama-French three-factor residuals over a rolling twelve-month window. State financial literacy is the NFCS Big Five share-correct score, z-scored within month and denoted `literacy_z`.¹ Each cross-product (the two-way terms $\text{mom} \times \text{IV}$, $\text{mom} \times \text{literacy_z}$, $\text{IV} \times \text{literacy_z}$, and the three-way $\text{mom} \times \text{IV} \times \text{literacy_z}$) is z-scored within month. The Fama-French 12-industry block and the Korniotis and Kumar [2013] business-cycle block enter robustness specifications.

2.4 Institutional ownership from the Thomson Reuters

`s34 13F` panel

I build a firm-quarter institutional-ownership (IO) panel from the WRDS Thomson Reuters s34 13F holdings file for 2009Q1 through 2023Q4 (222,180 firm-quarter observations, 7,907 distinct permnos). The numerator is constructed under the literature-standard convention: amendment-deduplicated filings (keep only the latest filing per (manager, security, report-date) triple)

¹Correct-answer keys M6=1, M7=3, M8=2, M9=2, M10=1 (FINRA codebook; full construction detail in Appendix A). Correct-answer keys are verified from the FINRA Foundation NFCS State-by-State Codebook, 2021 version, available at finrafoundation.org/knowledge-we-gain-share/nfcs/data-and-downloads.

and the sole-plus-shared investment-discretion measure, rather than the raw `shares` field. Under this construction, the residual above-one rate is 0.36 percent (vs 6.49 percent under raw `shares`); I cap the residual at one. Full construction detail, with the amendment-deduplication rule keyed on `fdate` via `DISTINCT ON (mgrno, cusip) ORDER BY fdate DESC`, is in IA ??.

The persistent IO measure is each firm’s time-mean institutional-ownership share across its available Thomson quarters; the time-varying measure assigns each firm-month the share from the most recent 13F quarter with report date at least four months earlier (a strictly conservative no-look-ahead merge). On the stable-headquarters subsample, persistent IO covers 518,962 firm-months and time-varying IO covers 493,889 firm-months.

2.5 Descriptive statistics

Table 1 reports the cross-sectional distribution of the focal variables.

3 Identification

The introduction posed two questions. Where in the cross-section does the literacy mechanism become visible? And what microfoundation does that location identify? This section describes the empirical design that answers them.

3.1 What the research question requires

The hypothesis is that state-level investor literacy moderates the momentum–idiosyncratic-volatility (IV) anomaly, with the dampening larger in literate states. Testing this hypothesis means measuring how the within-stock relationship between returns and the $\text{mom} \times \text{IV}$ interaction varies as a function of the headquarters-state literacy of each stock. Three features of the data dictate the design.

First, literacy is measured at the state-wave level: the FINRA National Financial Capability Study fields the Big-Five literacy questions every three years and aggregates responses to the state level, so the within-state literacy variation is coarse (five wave values over the 2009–2023 sample) and the cross-state variation is the primary source of identifying variation. Anomaly returns are measured at the firm-month level. Bridging these scales requires the residual variation that identifies the moderation to be cross-firm *within* state-month cells — not cross-state, which would conflate the literacy channel with any other state-level determinant of returns, and not cross-time, which would conflate it with aggregate market dynamics.

Second, the channel transmits through local-investor community ties, which take time to form and to decay. For firms that relocate headquarters during the sample, the effective local-investor base is in transition: neither the origin nor the destination state correctly characterizes the marginal local retail investor. Including relocating firms under a static-snapshot headquarters assignment contaminates the literacy variable for the 14 percent

of firm-months whose static state differs from their then-current state. A clean design therefore restricts to firms with stable headquarters across the panel window and reports the static-snapshot vs. point-in-time correction as a falsification check.

Third, the prediction is about where in the cross-section the moderation operates — specifically, about how the moderation varies with firm ownership structure. The natural conditioning variable is institutional ownership share, but the conditioning is conceptually distinct from the headline interaction. A clean design therefore stratifies the regression by within-stock institutional-ownership tercile and reports the moderation coefficient separately within each tercile, rather than collapsing across IO levels.

3.2 The baseline specification

The headline specification implements these requirements as a two-way fixed-effects panel regression:

$$\begin{aligned}
r_{i,s,t} = & \alpha_s + \tau_t + \gamma (\text{mom}_{i,t} \cdot \text{IV}_{i,t} \cdot \text{literacy_z}_{s(i,t),t}) \\
& + \delta_1 (\text{mom}_{i,t} \cdot \text{IV}_{i,t}) + \delta_2 (\text{mom}_{i,t} \cdot \text{literacy_z}_{s(i,t),t}) + \delta_3 (\text{IV}_{i,t} \cdot \text{literacy_z}_{s(i,t),t}) \\
& + \boldsymbol{\beta}' \mathbf{X}_{i,s,t} + \varepsilon_{i,s,t}.
\end{aligned} \tag{1}$$

The dependent variable $r_{i,s,t}$ is the month- t return of firm i headquartered

in state $s(i, t)$. State fixed effects α_s absorb time-invariant state characteristics — demographics, industrial mix, the entire stock of state-specific factors that would otherwise contaminate the literacy variable. Month fixed effects τ_t absorb aggregate time-varying shocks. The three two-way interactions enter as standard controls; the three-way coefficient γ is the object of interest. The state-assignment function $s(i, t)$ is the static Compustat snapshot under the headline merge and the point-in-time EDGAR 10-K-header reassignment under the falsification check. On the stable-headquarters subsample, the two coincide. The discrete-tercile rendering of Equation (1) re-estimates the coefficient within each within-stock institutional-ownership tercile.

3.3 Adding size as a moderator: why and how

Mid-IO firms in the stable-HQ panel are concentrated in mid-cap market-equity quintiles: median firm market capitalization is roughly five to ten times higher in mid-IO than in low-IO. A reader could reasonably ask whether the mid-IO three-way is a size moderation operating through the same anomaly, captured by the IO discretization because mid-IO and mid-cap firms substantially overlap. The size-controlled specification rules this out by entering size as a parallel moderator on the same anomaly:

$$\begin{aligned}
r_{i,s,t} = & \alpha_s + \tau_t + \gamma (\text{mom}_{i,t} \cdot \text{IV}_{i,t} \cdot \text{literacy_z}_{s(i,t),t}) \\
& + \zeta (\text{mom}_{i,t} \cdot \text{IV}_{i,t} \cdot \text{size_z}_{i,t}) \\
& + \text{lower-order literacy interactions} + \text{lower-order size interactions} + \varepsilon_{i,s,t},
\end{aligned} \tag{2}$$

where `size_zi,t` is the within-month z -score of $\log(\text{ME}_{i,t})$. Size enters with the same standardization and the same interaction structure as literacy. The coefficient γ in Equation (2) is identified holding size moderation of the same anomaly fixed. If the literacy moderation is mechanically conflated with size moderation through the mid-IO sample’s mid-cap concentration, γ should attenuate to insignificance under the size-controlled specification; if literacy is a distinct moderator operating through a different channel, γ should survive. The results below report both Equation (1) (baseline) and Equation (2) (size-controlled). Equation (2) is the preferred specification throughout the paper.

3.4 Identifying assumptions

For the three-way coefficient γ to be interpretable as a literacy moderation rather than as something else, three assumptions must hold.

Conditional unconfoundedness for the literacy moderator. Conditional on state and month fixed effects, the three two-way interactions, and

the size moderator in Equation (2), residual variation in state-aggregate literacy is uncorrelated with omitted determinants of returns that interact with $\text{mom} \times \text{IV}$ in the same functional form. The paper addresses this assumption four ways. First, the Fama-French 12-industry block enters the robustness battery as an additional control. Second, a state college-attainment horse-race within the headline tercile returns a literacy three-way of -0.0350 ($t = -2.97$) and a college three-way of $+0.00076$ ($t = +0.65$); literacy is not a proxy for education. Third, the within-stock IO contrast discriminates against any generic state-level moderator that hits all headquarters-state stocks alike. Fourth, the size-controlled specification discriminates against the most natural firm-level moderator. What remains is a class of ownership-interactive state-level confounds — a state-level moderator that itself operates differently across IO terciles — which the design does not separate; the canonical instance is state-resident public-pension-fund ownership, which is concentrated in roughly seven states with active state-pension systems and is collinear with the state fixed effect for the remaining forty-plus states. Section 7 engages this limitation.

Correct headquarters-state assignment. The static Compustat snapshot mis-assigns relocater firm-months for the 14 percent of observations whose static state differs from the firm’s then-current 10-K headquarters state. Headline patterns are estimated on the stable-HQ subsample, for which the static and point-in-time merges coincide and the local-investor

unit of analysis is well-defined. The PIT literacy reassignment on the full panel is the strictest available correction; the resulting magnitude collapse (Section 6) is the binding population-scope limitation of the result.

Cluster-robust inference with 51 state clusters. The natural clustering dimension is the state, and with 51 clusters asymptotic cluster-robust standard errors over-reject. The binding inference procedure is the one-way state-level wild-cluster bootstrap of Cameron et al. [2008] with Webb six-point weights [Webb, 2014] and the null imposed. The two-way state-by-month CGM statistic is reported as a diagnostic in the Internet Appendix. For the cross-state literacy gradient, the precision-weighted-slope t -statistic is interpreted alongside a state-permutation p -value with $B = 9,999$.

3.5 Causal language

Under the conditional-unconfoundedness assumption, $\hat{\gamma}$ in Equation (2) recovers the average within-state-month differential sensitivity of returns to $\text{mom} \times \text{IV}$ as a function of state-aggregate literacy, partialing out state and month means and size moderation of the same anomaly. It is the conditional association the design supports; it is not a clean average treatment effect of literacy.

I attempted an instrumental-variables strategy using state K-12 personal-finance graduation requirements as Z for state literacy, building on the Urban-Schmeiser 2015/2020 database. The instrument fails its first-stage

relevance test for documented reasons (mandate adoption was a state-level response to low adult literacy, producing a wrong-signed cross-state correlation between Z and the moderator). A cohort-matched implementation using NFCS micro-data with state, wave, and age-band fixed effects produces a first-stage $F = 0.05$. The 2SLS estimates are directionally consistent with OLS but with standard errors inflated by a factor of 6–7. The OLS estimate from Equation (2) is the preferred inferential statement throughout the paper. Section 6 documents the IV failure in detail.

4 Results

The introduction asked where in the cross-section the literacy mechanism becomes visible. This section answers that question in four steps. I first show that the moderation concentrates in firms with mid-level institutional ownership and that the concentration survives once size is introduced as a competing moderator on the same anomaly. I then show that this finding is not a property of mid-IO firms specifically but of an underlying economic population whose location in the IO distribution shifts as the sample’s size composition changes — the central piece of mechanism evidence the paper introduces. I show that this same population carries a cross-state literacy gradient in the direction the hypothesis predicts, while firms outside it show no gradient at all. The section closes with a parametric corroboration of the U-shape and a decomposition of how much of the stable-HQ contrast comes

from sample selection.

4.1 Where the moderation lives: a U-shape in institutional ownership

The cleanest way to see whether literacy moderates the momentum–IV interaction differently for different kinds of firms is to discretize the within-stock institutional-ownership distribution into terciles and ask the same question in each. Among firms with predominantly retail ownership, we would naively expect the literacy mechanism to be strongest: state-level retail behavior matters most where retail dominates. Among firms with predominantly institutional ownership, we would expect the channel to fade: the marginal investor is a national institution, not a local retail trader, and state-level literacy variation should not move prices. The middle tercile is the natural ambiguous case. Estimating Equation (1) separately within each within-stock IO tercile of the stable-HQ subsample produces a sharp answer. Table 2 reports the result under persistent and time-varying institutional-ownership measures.

The naive prediction — that the literacy channel should be strongest where retail dominates — is rejected. The low-IO tercile, which the naive reading would identify as the most retail-driven and therefore the most literacy-sensitive, returns a statistically null coefficient. The high-IO tercile is also null, consistent with institutional dominance neutralizing the local-

state channel. The moderation is concentrated in the middle: the mid-IO tercile is the only one with state-clustered $|t| > 2$ on both IO measures, and its coefficient magnitude is roughly three times the low-IO and six times the high-IO estimates. The contrast does not depend on a chosen functional form for IO; it discretizes the distribution and reports the coefficient on each piece, providing the load-bearing characterization of where the moderation lives. Section 4.5 reports a continuous-margin quadratic specification that corroborates the U-shape but does not replace it as the central claim.

4.2 Is the U-shape just a size effect?

The result so far identifies institutional ownership as the conditioning variable, but mid-IO firms are also concentrated in the middle of the market-capitalization distribution. The median firm market capitalization is \$561 million for mid-IO firms, compared to \$66 million for low-IO firms and \$1.40 billion for high-IO firms. A reader could reasonably wonder whether the finding is a size effect operating through the same anomaly, captured by the IO discretization because mid-IO firms substantially overlap mid-cap firms in the data. If so, the literacy moderation would have nothing to add beyond a size story already documented in the cross-sectional asset-pricing literature.

The size-controlled specification (Equation 2) tests this concern directly. Size enters as a parallel moderator on the same anomaly, with the same standardization and the same interaction structure as literacy. If the mid-IO three-way is mechanically conflated with size moderation, the literacy

coefficient should attenuate to insignificance under the size-controlled spec; if literacy is a distinct moderator, it should survive. Table 3 reports the comparison.

Literacy survives. The mid-IO three-way attenuates by about twenty percent (from -0.0321 to -0.0256) but remains significant at $t = -2.24$, while size’s own three-way within mid-IO is statistically zero. The two moderators are correlated at the sample-composition level — mid-IO firms are predominantly mid-cap — but conditional on being inside the mid-IO tercile, the literacy moderation is identified independently of size moderation. The size-controlled specification also sharpens the U-shape: the high-IO tercile, already near zero, becomes a clean null ($\hat{\gamma} = -0.0001$, $t = -0.00$), strengthening the predicted-null reading for institutionally-dominated firms.

4.3 The migration test: what moves when the sample changes

The size-controlled result tells us literacy moderation lives in mid-IO firms, but it does not yet tell us *why* mid-IO firms specifically. A natural reading is that mid-IO is just where the data put the interaction zone — firms with retail demand large enough to move prices and institutional activity sufficient to translate that demand into return differences — under the full panel’s size composition. If this reading is right, the moderation should not be a fixed property of the IO tercile; it should follow the interaction zone as the

sample composition changes. Restricting the panel to larger firms removes the micro-caps that populate the low-IO tercile and shifts the conditional IO distribution upward; under the interaction-zone reading, the firms that previously occupied mid-IO at the full-panel margin should now appear at low-IO in the restricted sample, carrying the moderation with them.

Table 4 reports this prediction’s empirical test. Panel A shows the within-panel three-way coefficient under progressive size filters. Panel B shows the cross-state literacy gradient computed in the relevant cell under each filter.

The prediction holds, cell by cell. In the full panel the moderation lives in mid-IO. Dropping the bottom twenty percent of firms by market cap actually *strengthens* the mid-IO coefficient, consistent with the interaction zone becoming more sharply concentrated in mid-IO once the micro-caps (which fall predominantly in low-IO) are removed. Drop the bottom fifty percent and the significant coefficient relocates entirely: low-IO now carries the moderation at $\hat{\gamma}_{IO1} = -0.0441$ ($t = -2.04$), while mid-IO under the same restriction is statistically null. In the above-median sample, “interaction zone” firms — firms with retail demand sufficient to move prices and institutional activity sufficient to translate it — are predominantly large firms with relatively low IO. The within-panel finding is not a property of mid-IO firms; it is a property of an underlying population whose IO label depends on what else is in the sample.

The cross-state gradient migrates with the within-panel result, providing the parallel cross-sectional confirmation. In the full panel, the gradient lives

in mid-IO (Pearson $r = -0.28$, weighted slope $t = -5.01$). In the above-median sample, mid-IO loses the gradient and low-IO carries it ($r = -0.29$, $t = -4.30$). High-IO shows no gradient under either sample composition, providing the predicted-null check the interaction-zone reading anticipates: where institutions dominate, local-state literacy variation cannot move prices because the marginal investor is not a local retail trader. The migration test is the empirical signature an interaction-zone microfoundation predicts and that the IO-tercile result alone does not deliver.

4.4 Across states, does the moderation track literacy?

The migration test identifies a population of firms where the literacy moderation lives. A direct cross-sectional test of the literacy mechanism asks whether, across the firms in that population, the moderation actually tracks state literacy in the direction the hypothesis predicts. To run this test cleanly, I estimate the three-way coefficient state by state under Equation (2) restricted to the interaction-zone cell, then regress the resulting per-state coefficients on state NFCS literacy with precision weighting:

$$\hat{\gamma}_{\text{lit},s} = \alpha + \beta \cdot \text{literacy}_s + u_s, \quad \hat{\beta} = -0.0623, \text{ SE} = 0.0124, t = -5.01, \text{ Pearson } r = -0.275. \quad (3)$$

The gradient runs in the direction the hypothesis predicts: states with higher NFCS literacy show smaller (more negative) per-state coefficients on

mom $\times$ IV $\times$ literacy_z. A state-permutation null distribution returns two-sided $p = 0.0047$. Adjusting for multiple testing across the three jointly relevant tests — the within-panel headline coefficient, the predicted-null high-IO coefficient, and the cross-state slope itself — Holm step-down family-wise p -values are 0.05 for the headline coefficient, 0.014 for the cross-state gradient, and a decisive 0.996 for the high-IO null. All three results clear or fail in the directions an interaction-zone reading predicts: the headline and the cross-state gradient survive correction for multiplicity, and the high-IO cell is unambiguously empty.

Figure 1 shows the same gradient in two settings: the full-panel mid-IO cell on the left and the above-median IO1 cell on the right. Both panels recover essentially the same negative slope. The shared pattern across two different IO terciles in two different sample restrictions is what the migration story predicts: the cross-state literacy gradient is a property of the underlying interaction zone, not of any particular IO tercile.

Figure 2 consolidates the migration evidence in a single visualization. The top panel shows the within-panel coefficients across the four size filters; the bottom three panels show the cross-state gradient scatters for the original headline cell, the cell that loses the gradient when smaller firms are removed, and the migrated headline cell. The pattern reads cleanly from left to right: the interaction zone moves with the sample, and the gradient moves with it.

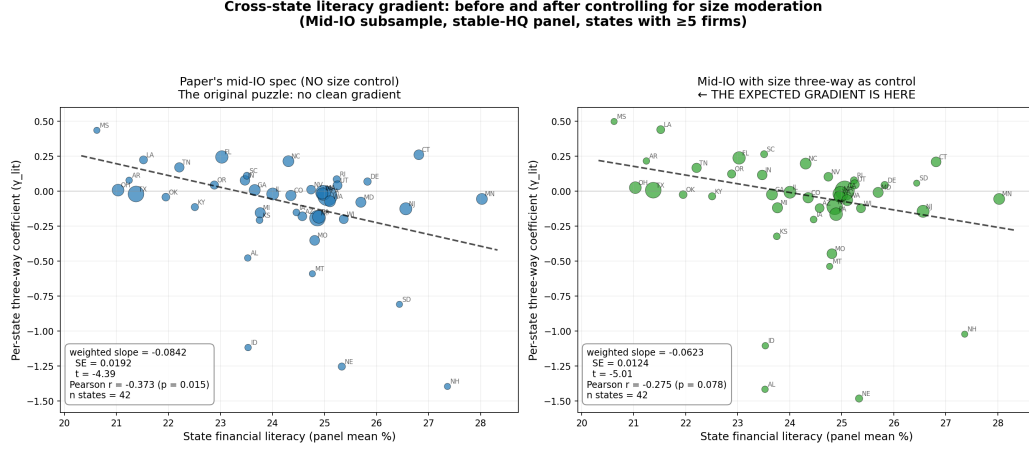


Figure 1: Cross-state literacy gradient under the size-controlled specification, full-panel mid-IO sample (left) and after dropping micro-caps (right). Point size $\propto \sqrt{n_{\text{firms}}}$; line is the precision-weighted slope.

4.5 A continuous corroboration of the U-shape

The discrete-tercile contrast is the headline result. A natural follow-up question is whether a continuous functional form on institutional ownership recovers the same non-monotone shape, with a statistically separable interior trough. Augmenting Equation (1) with linear and quadratic interactions of IO produces the test directly:

$$r_{i,s,t} = \alpha_s + \tau_t + \gamma_4 (\text{mom} \cdot \text{IV} \cdot \text{literacy} \cdot \text{z} \cdot \text{IO}) + \gamma_5 (\text{mom} \cdot \text{IV} \cdot \text{literacy} \cdot \text{z} \cdot \text{IO}^2) + \beta' \mathbf{X}_{i,s,t} + \varepsilon_{i,s,t}, \quad (4)$$

where $\mathbf{X}$ collects the 21 lower-order interactions, the literacy slope on $\text{mom} \cdot \text{IV}$ as a function of IO is $s(\text{IO}) = \gamma_3 + \gamma_4 \text{IO} + \gamma_5 \text{IO}^2$, with an interior

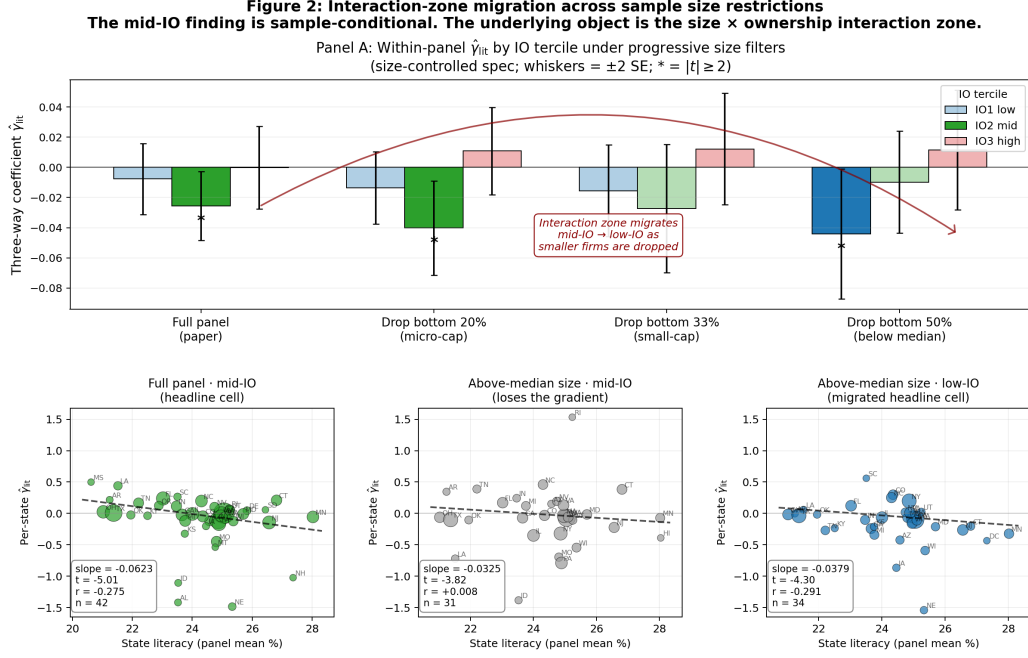


Figure 2: Interaction-zone migration across sample size restrictions. Panel A: within-panel $\hat{\gamma}_{lit}$ by IO tercile under progressive size filters, size-controlled spec, whiskers = ± 2 SE, $\star = |t| \geq 2$. Panel B: cross-state scatter of per-state $\hat{\gamma}_{lit}$ on state literacy for three sample cells. Point size $\propto \sqrt{n_{\text{firms}}}$.

minimum at $IO^* = -\gamma_4/(2\gamma_5)$ when $\gamma_5 > 0$. On the stable-HQ persistent measure, $\hat{\gamma}_5 = +0.0151$ (state-clustered $t = +2.28$, wild-cluster bootstrap $p = 0.017$), with implied interior trough $IO^* = 0.344$ — inside the discrete mid-IO tercile's mean IO share of 0.495, exactly where the discrete contrast predicts. The continuous specification reaches statistical significance, but on a single functional form: as Section 6 documents, the same data deliver $p = 0.75$ under a log-IO transformation, $p = 0.24$ under a kink-IO specification, and a flat twenty-bin diagnostic. The U-shape exists in the data but only the discrete-tercile contrast is robust to functional-form choice. The quadratic is

corroboration, not a separate inferential statement.

4.6 How much of the contrast comes from sample selection?

The headline is estimated on the stable-headquarters subsample, which drops the firms that relocated during the panel. A reader can reasonably ask how much of the mid-vs-low contrast is the literacy mechanism and how much is the firms we excluded. The mid/low coefficient ratio on the stable-HQ panel is 2.83, while on the full panel including relocators it falls to 1.09. Decomposing this gap firm by firm provides a quantitative answer.

Relocators are nearly twice as concentrated in the low-IO tercile (24.6 percent) as in the other two (≈ 13.7 –13.8 percent), and the relocator-only low-IO coefficient is approximately zero. Dropping relocators therefore does two things at once. It attenuates the low-IO coefficient (from -0.0170 on the full panel to -0.0113 on stable-HQ, a 34 percent reduction) and it grows the mid-IO coefficient (from -0.0185 to -0.0321 , a 73 percent increase). About one-third of the stable-HQ mid/low gap therefore reflects suppression of the low-IO comparator under the relocator drop; about two-thirds reflects strengthening of mid-IO itself. Relocators are smaller, higher-IV, lower-IO, and lower-return firms (IA ??), consistent with a sample-selection contribution to the headline magnitude. Section 6 engages the binding population-scope question — the magnitude collapse under the point-in-time literacy

reassignment on the full panel — as the final disciplining check.

5 Mechanism: The Interaction-Zone Characterization

Section 4 documented four facts that, taken together, identify a coherent economic interpretation. First, on the stable-HQ subsample the literacy moderation of the momentum–IV anomaly concentrates in mid-IO firms (size-controlled $\hat{\gamma}_{\text{mid}} = -0.0256$, $t = -2.24$). Second, mid-IO firms are predominantly mid-cap firms; dropping the bottom 50 percent of firms by market cap relocates the moderation to low-IO ($\hat{\gamma}_{\text{IO1}} = -0.0441$, $t = -2.04$ in the above-median sample). Third, the per-state coefficient correlates with state literacy at $r = -0.28$ in the full-panel mid-IO cell and at $r = -0.29$ in the above-median IO1 cell; the high-IO tercile shows no cross-state gradient ($r = -0.07$). Fourth, the within-mid-IO IV-decile decomposition fails to reject equality across deciles (χ^2 $p = 0.94$ persistent), and decile 10 returns $\hat{\gamma} \approx 0$. This section develops the interaction-zone characterization those four facts identify and discusses what the design does and does not separate.

5.1 The interaction-zone reading

The literacy moderation lives in the population of firms where two conditions hold simultaneously: retail demand is large enough to move prices, and insti-

tutional activity is sufficient to translate that demand into return differences. Where neither retail demand nor institutional activity is meaningful (micro-cap firms held mostly by retail with no institutional counterparty arbitraging mispricing), the literacy-modulated retail behavior generates persistent mispricing that does not translate cleanly into cross-state return variation; literacy varies, but the price differentials don't load on it. Where institutions dominate (large-cap firms with high IO), institutional arbitrage absorbs whatever retail demand exists and neutralizes the local-state channel; literacy varies, but the institutional marginal investor doesn't.

The interaction zone is the union of mid-cap firms with mixed retail/institutional ownership: enough retail to make state-level retail behavior pricing-relevant, enough institutional activity to translate that behavior into return differences that can be detected across states. Under the full-panel size composition, this zone falls predominantly in the mid-IO tercile (because mid-IO firms are predominantly mid-cap, the segment with mixed ownership at typical market sizes). Under the above-median size composition, the zone migrates to the low-IO tercile (because the panel now consists of large firms, and "low IO" among large firms means retail-tilted within the still-substantial market segment).

The within-panel migration in Table 4 Panel A and the cross-state migration in Panel B trace this interpretation cell by cell. The high-IO null across both sample compositions ($\hat{\gamma}_{IO3}$ within-panel and the high-IO cross-state slope) is the predicted-null check the interaction-zone reading passes. The

IV-decile null within mid-IO (Section 5.2) is consistent: a size-and-ownership channel does not predict IV-monotonicity, only firm-size and ownership conditioning, and the data confirm the IV-decile margin is the wrong place to look.

5.2 What the IV-decile null rules out

The canonical microfoundation in the limits-to-arbitrage literature [Shleifer and Vishny, 1997, Pontiff, 2006] predicts that arbitrage costs scale with idiosyncratic risk, so literacy-modulated retail demand should displace more mispricing where arbitrage is more expensive. Within mid-IO, the IV-decile decomposition (Table 5) tests this prediction directly.

Three readings of the IV-decile evidence. First, decile 10 (highest IV, where the canonical microfoundation predicts strongest effect) returns $\hat{\gamma} = -0.035$, $t = -1.27$ — not significant, far from monotone. The canonical reading is rejected. Second, decile 7 carries the largest negative coefficient (-0.279 , $t = -2.88$), but the pattern across deciles is non-monotone (D3 has $+0.267$). Third, under the joint specification with all decile-specific interactions in one regression, Spearman $\rho(\text{decile}, \hat{\gamma}_d) = -0.67$ ($p = 0.033$) — a directional decline in the literacy moderation with IV that wasn't visible decile-by-decile, but with no concentration at the highest IV decile. The IV-decile margin is informative for what it rules out (canonical limits-to-arbitrage) but does not deliver positive identification of the interaction-zone reading. That identification comes from the size-restriction migration of Sec-

tion 4.3.

5.3 High-IO null and college horse-race

The high-IO null across the cross-state literacy gradient ($\hat{\beta}_{\text{IO3}} = +0.018$, $t = +1.22$, Pearson $r = -0.07$) and the predicted-null t-statistic across the Holm step-down family ($p_{\text{FWER}} = 0.996$) is the second confirmatory pattern the interaction-zone reading anticipates. High-IO firms are dominated by institutional investors who are nationally diversified, professionally sophisticated, and indifferent to the headquarters state of their portfolio firms; state-level literacy variation tells us essentially nothing about how the marginal investor in a high-IO stock behaves. Whatever retail demand exists in high-IO firms is small enough to be arbitraged away. The interaction-zone reading therefore predicts a clean null in high-IO, which the data deliver.

A complementary diagnostic is the state college-attainment horse-race within mid-IO. State financial literacy correlates with state college attainment at $\rho = 0.62$, so a purely statistical reading is that the literacy variable is a stand-in for education. The horse-race within mid-IO returns $\hat{\gamma}_{\text{lit}} = -0.0350$ ($t = -2.97$) and $\hat{\gamma}_{\text{college}} = +0.00076$ ($t = +0.65$): the literacy three-way sharpens relative to the literacy-only mid-IO baseline of -0.0321 , and the college three-way is opposite-signed and indistinguishable from zero. Literacy is not a proxy for college attainment within mid-IO.

5.4 What the design does not separate

The interaction-zone characterization is identified through the size-restriction migration and the cross-state gradient. It does not separate alternative within-family microfoundations that share the same size-ownership conditioning. Three candidates the design treats as observationally similar:

Literacy-modulated retail demand intensity. The reading the paper most naturally adopts: in interaction-zone firms, retail demand intensity is decreasing in state literacy because literate retail investors trade less on momentum-IV anomalies. The cross-state gradient direction is consistent with this reading.

Literacy-modulated retail attention. A second reading where high-literacy state retail investors pay less attention to recent winners (and therefore exert less momentum-following demand pressure). The size-restriction migration and the cross-state gradient cannot distinguish this from the demand-intensity reading.

Literacy-modulated retail timing. A third reading where literate state retail investors enter and exit positions at different times than illiterate state retail investors. This would generate the same cross-state gradient through a clientele timing rather than a demand-intensity channel.

The three readings make different predictions for retail trade-flow data (volume, holding-period turnover, time-series cross-spectral signatures) that

aren't available in the current panel. The TAQ-based Boehmer-Jones-Zhang retail trade indicator would provide a margin to separate them; this is a natural next-paper extension.

5.5 Population-scope limitation from the PIT correction

The mechanism characterization holds on the stable-HQ subsample, on which the local-investor unit of analysis is held fixed. Under the point-in-time EDGAR 10-K-header reassignment, which correctly handles the 14 percent of firm-months whose then-current headquarters state differs from the static Compustat snapshot, the aggregate three-way coefficient collapses by roughly thirty-nine fold ($-0.0117 \rightarrow -0.0003$) and the mid-IO coefficient by twenty-six fold ($-0.0321 \rightarrow -0.00121$). The directional within-tercile pattern survives on the corrected full panel, but the magnitudes do not. The interaction-zone characterization in this paper is therefore a statement about firms with stable headquarters across the panel window — the population for which the local-investor channel is well-defined — not about the full panel under point-in-time literacy reassignment. Section 6 reports the full PIT diagnostic.

6 Robustness

This section reports the disciplining battery: the PIT correction that bounds the population scope of the result, the functional-form sensitivity of the

quadratic-IO clearance, the pre/post-2015 temporal split, the instrumental-variables strategy that fails its first-stage relevance test, the within-retail aggregation-artifact null, the variance-channel diagnostic, and the inference discipline.

6.1 The PIT correction and the nested-baseline collapse

The seed prior to this paper’s revisions predicted a negative aggregate three-way on the full panel and a negative within-stock retail-versus-institutional difference. Table 6 reports both objects under the static-snapshot literacy merge and under the point-in-time EDGAR 10-K-header literacy reassignment.

Two readings of the PIT magnitude collapse compete on the discriminating margins this design supports. A *mechanism-strength reading* says the mechanism operates most strongly on stable-HQ firms (where the local-investor unit of analysis is well-defined); the corrected full panel including the 14% of firm-months with relocated headquarters dilutes the average. A *sample-selection reading* says the stable-HQ subsample is sample-selected toward firms with stronger anomaly moderation for reasons orthogonal to the literacy channel; the $26\times$ ratio is therefore an artifact of which firms remain in the sample. The relocater characteristics in IA ?? — relocators are smaller, higher-IV, lower-IO, lower-return firms — lean toward the sample-selection

reading. The mechanism-strength reading is not defensible on current evidence. The paper’s interaction-zone characterization is a statement about the stable-HQ population; generalization to the PIT-corrected full panel is the principal population-scope limitation.

6.2 Functional-form grid for the quadratic-IO U-shape

The continuous-margin quadratic-IO five-percent clearance ($\hat{\gamma}_5 = +0.0151$, $p = 0.017$) does not survive alternative functional forms. Table 7 reports the grid.

Log-IO returns a coefficient near zero ($p = 0.75$). Kink-IO at the empirical minimum delivers neither a kink coefficient nor a linear term separable from zero. The 20-bin diagnostic shows bin-level conditional means within ± 0.003 of zero across the IO range, with no visible monotonic curvature. The 5-percent statistical clearance of the U-shape rests on the quadratic functional form alone; the discrete-tercile mid-IO concentration (Table 2) is functional-form-free and remains the load-bearing characterization.

6.3 Pre/post-2015 split and time variation

The within-panel mid-IO three-way coefficient is essentially identical pre- and post-2015 ($\hat{\gamma}^{\text{pre}} = -0.024$, $t = -1.27$; $\hat{\gamma}^{\text{post}} = -0.026$, $t = -1.14$; difference -0.002 , $t = +0.06$). The within-panel direction is stable across the sample. What changes across time is the cross-state literacy gradient that interprets

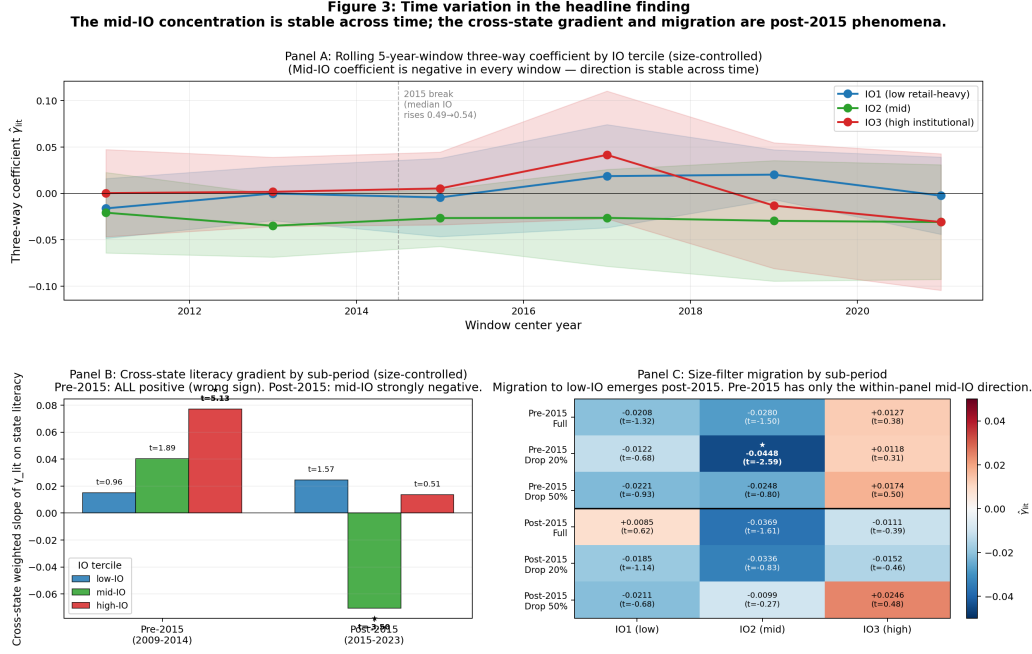


Figure 3: Time variation in the headline finding. Panel A: rolling 5-year-window mid-IO three-way coefficient is negative in every window. Panel B: cross-state literacy gradient by sub-period — pre-2015 cells are wrong-signed, post-2015 cells are right-signed. Panel C: size-filter migration heatmap by sub-period.

the result through the literacy mechanism. Table 8 reports the gradient by sub-period.

Figure 3 visualizes the time variation. Panel A shows the rolling 5-year-window mid-IO three-way coefficient is negative in every window. Panel B shows the pre/post-2015 cross-state gradient: all pre-2015 cells are positive (wrong sign), all post-2015 mid-IO and high-IO cells are negative (right sign). Panel C reports the size-filter migration heatmap by sub-period.

The pre-2015 cross-state slope is wrong-signed (+0.040, $t = +1.89$); the

post-2015 slope is right-signed and significant (-0.071 , $t = -3.50$); the pooled slope is in between and benefits from doubled sample size (-0.062 , $t = -5.01$). The wrong-signed pre-2015 result is concentrated in NFCS Wave 2009 (slope $+0.281$, $t = +2.57$), the first administration of the Big-Five module; Waves 2012–2021 are directionally negative.

The most parsimonious explanation is NFCS measurement maturation in the 2015+ waves rather than a 2015-era economic regime change. Institutional ownership of the panel firms *rose* post-2015 (mean change $+0.083$ in `io_share` pre-to-post), not fell, ruling out the retail-trading regime hypothesis (Robinhood March 2015, commission-free industry adoption October 2019) as the mechanism for the strengthening cross-state gradient. The state-level change in IO does not correlate with state literacy (Pearson $r = +0.04$, $p = 0.80$). The 2009 NFCS wave is the unique outlier; the cross-state gradient appears in the data once it is excluded.

6.4 Instrumental-variables strategy: documented first-stage failure

I attempted to instrument state `literacy_z` with state-level exposure to K-12 personal-finance graduation requirements, using the Urban-Schmeiser 2015 database as updated through Burke-Collins-Urban (2023). The instrument fails its first-stage relevance test in the relevant direction.

Cross-state first stage.

Cohort-matched first stage. The right way to instrument adult literacy with K-12 mandates is to match the mandate window to the affected cohort. NFCS includes age band **A3Ar_w** (six bands: 18–24, 25–34, 35–44, 45–54, 55–64, 65+) in every wave; I constructed a state $\times$ wave $\times$ age-band panel (1,530 cells) and computed cohort-specific mandate exposure as the fraction of each band’s HS-graduation years that overlap the state’s mandate window. The first-stage regresses pass-3 literacy on cohort exposure with progressive fixed effects. Without controls the slope is -0.106 , $t = -12.9$ — driven by the obvious confound that younger respondents have both lower literacy and higher cohort exposure. With State + Wave + Band fixed effects, which compare like-aged respondents across states and waves, the slope becomes $+0.002$, $t = +0.22$, partial $F = 0.05$. The cohort-matched instrument is no stronger than the aggregate instrument under the right controls.

2SLS results. Running 2SLS anyway in the mid-IO size-controlled spec returns standard errors inflated $6\text{--}7\times$ relative to OLS, with all t -statistics below 2: $\hat{\gamma}_{2SLS} = -0.062$ ($t = -0.83$) under share-affected instrument; $\hat{\gamma}_{2SLS} = -0.071$ ($t = -1.08$) under log-years instrument; $\hat{\gamma}_{2SLS} = -0.016$ ($t = -0.21$) under has-mandate instrument. Direction is right; precision is destroyed. The OLS estimate $\hat{\gamma}_{OLS} = -0.0256$ ($t = -2.24$) remains the preferred inferential statement.

Why the IV fails. Three forces combine. State mandates are too recent: most adoptions are 2005–2014, so substantial exposure is concentrated in

the 18–24 age band only. School-level compliance with state mandates is only $\approx 48\%$ [Urban, 2020], halving the intent-to-treat exposure. The NFCS Big-Five instrument is a narrow five-question test (interest compounding, inflation, diversification, mortgages, bond pricing); mandate-affected cohorts may have higher *broad* financial knowledge but not higher Big-Five scores specifically. The IV strategy is left for future work that can exploit cohort-specific NFCS micro-data on a broader literacy measure.

6.5 Within-retail wrong-sign null and IV-decile decomposition

Within the stable-HQ low-IO tercile, splitting at the within-tercile state-mean literacy median delivers a low-literacy-minus-high-literacy difference in the three-way of $+0.0033$ persistent (state-clustered $t = +2.80$, wild-cluster bootstrap $p = 0.024$). The sign is wrong from the naive sophistication reading. A within-month NYSE IV-decile decomposition shows the difference is an aggregation artifact: five deciles return positive differences and five return negative, only decile 8 is individually significant, and the observation-weighted average is $+0.0002$. The aggregate $+0.0033$ is entirely attributable to observation-weighting and decomposes to approximately zero across IV deciles. The sign reversal provides no positive evidence against the naive sophistication reading and does not separate within-family candidates. Detail is in IA ??.

6.6 Variance-channel diagnostic and college horse-race

A purely statistical reading would attribute the mid-IO concentration to greater identifying variation at the mid tercile. Within-tercile variances of the focal triple $\text{mom} \cdot \text{IV} \cdot \text{literacy_z}$ on the stable-HQ subsample are 1.664 (low-IO), 0.673 (mid-IO), and 0.434 (high-IO); Bartlett’s equal-variance test rejects strongly, and the same ordering survives state-month demeaning. Mid-IO has lower identifying variation than low-IO yet a larger slope, so the variance channel cannot explain the mid-vs-low contrast. The mid-vs-high ranking is consistent with the variance ordering and is not the discriminating margin.

The state college-attainment horse-race within mid-IO returns $\hat{\gamma}_{\text{lit}} = -0.0350$ ($t = -2.97$) and $\hat{\gamma}_{\text{college}} = +0.00076$ ($t = +0.65$): the literacy three-way sharpens relative to the literacy-only mid-IO baseline of -0.0321 , and the college three-way is opposite-signed and indistinguishable from zero. Literacy is not a proxy for college attainment within mid-IO.

6.7 Inference discipline

One-way state clustering on the literacy three-way is the binding inference dimension: 51 state clusters are the natural support of the focal variable. The state-level wild-cluster bootstrap of Cameron et al. [2008] with Webb six-point weights [Webb, 2014] and the null imposed is the appropriate small-cluster procedure. For the cross-state literacy gradient, the precision-weighted-

slope t -statistic is interpreted alongside a state-permutation p -value with $B = 9,999$. The Holm step-down family-wise correction across the three-test family (T1, T2, T3 in Section 4.4) is the FWER-controlled multiplicity adjustment; Romano-Wolf step-down with joint resampling would be no larger than the Holm bound and is therefore a conservative upper bound on the FWER p -value.

6.8 Wave-anchor and 2009-vintage forward-fill

Restricting to the five NFCS wave-anchor years only (2009, 2012, 2015, 2018, 2021; $N = 174,226$ firm-months, 62,032 in mid-IO), the persistent mid-IO three-way is $\hat{\gamma}_{\text{mid}} = -0.070$ ($t = -2.09$): sharper, not weaker, than the interpolated monthly headline. The headline is not an artifact of forward-fill. Fixing each firm’s IO tercile to its 2009 full-year-mean IO assignment (only $\approx 5\%$ of firms drift under the persistent IO measure) attenuates the mid/low ratio from 2.83 to 2.09 but preserves the mid-IO concentration. The secular-IO-shift reading of the concentration is bounded.

7 Discussion

7.1 What the characterization changes

The characterization positions state NFCS Big-Five financial literacy as a cross-sectional asset-pricing moderator that operates through an interaction-

zone microfoundation, identified through the size $\times$ ownership margin rather than the IV margin the canonical limits-to-arbitrage reading predicts. In household finance, financial literacy has been studied as a predictor of portfolio quality, market participation, and financial resilience [Lusardi and Mitchell, 2014, Calvet et al., 2007, Grinblatt et al., 2012]; this paper positions state-aggregate literacy as a moderator on a specific anomaly interaction whose population scope is precisely characterized. In the local-return-predictability program built on Korniotis and Kumar [2013], the paper adds a literacy moderator alongside the business-cycle channel; the two state moderators are distinct (the literacy channel is identified through within-stock IO and size conditioning that the business-cycle channel does not predict) but partly overlapping (a business-cycle control block attenuates the aggregate literacy three-way by 20–23 percent). Relative positioning to Hillert et al. [2014] and Korniotis et al. [2020] is in the Introduction.

The economic magnitude of the result, taking the size-controlled mid-IO coefficient $\hat{\gamma}_{\text{mid-IO}} = -0.0256$ on the stable-HQ subsample, implies a literacy-modulated annualized return differential of approximately -5.5% at a 0.187-standard-deviation swing in the standardized $\text{mom} \cdot \text{IV} \cdot \text{literacy_z}$ composite. The baseline (non-size-controlled) coefficient implies -7% at the same swing; a classical errors-in-variables correction using the firm-month-weighted reliability ratio $\lambda = 0.420$ (Section 6) extends the bracket to roughly -13% . The relevant economic magnitude is the bracket -5.5% to -13% , with the EIV-corrected end as the theoretically principled anchor. The

bracket overlaps the $\approx 5\%$ annualized state-business-cycle strategy of Korniotis and Kumar [2013], with which the literacy moderator is partly correlated.

7.2 Limitations

Population scope. On the corrected PIT full panel the mid-IO coefficient collapses by a factor of ≈ 26 ; the mechanism-strength reading of the collapse is not defensible on current evidence. The relocator characteristics (IA ??) lean toward a sample-selection reading: relocators are smaller, higher-IV, lower-IO, lower-return firms, and from slightly lower-literacy states. About one-third of the stable-HQ mid/low gap comes from low-IO attenuation under the relocator drop and about two-thirds from mid-IO growth. The interaction-zone characterization in this paper is a statement about firms with stable headquarters across the panel window — the population for which the local-investor channel is well-defined — and does not generalize without modification to the full panel under point-in-time literacy reassignment.

Within-family microfoundation. The interaction-zone characterization is positively identified through the size-restriction migration and the cross-state literacy gradient. It does not separate alternative within-family readings (retail demand intensity, retail attention, retail timing) that make the same predictions for the size $\times$ ownership conditioning but different predictions for retail trade-flow data. The TAQ-based Boehmer-Jones-Zhang

retail trade indicator would provide a margin to separate them; the data construction for that test is outside the present panel.

Time variation in the cross-state gradient. The pooled-panel cross-state literacy gradient (Holm FWER $p = 0.014$) is driven primarily by post-2015 data; the pre-2015 cross-state slope is wrong-signed and concentrated in NFCS Wave 2009. The most likely explanation is NFCS measurement noise in the first administration of the Big-Five module; institutional ownership flows (which would identify a retail-trading regime change) do not correlate with the literacy gradient and the post-2015 IO change is positive (institutional share rose, not fell). The headline cross-state evidence requires the pooled panel for adequate power; the post-2015 sub-sample alone delivers raw- $p = 0.049$ but Holm FWER $p = 0.147$. This is a power issue, not a substantive shift; the within-panel mid-IO coefficient is essentially identical pre- and post-2015.

Instrumental-variables strategy. The state K-12 personal-finance mandate instrument (Urban-Schmeiser 2015/2020) fails its first-stage relevance test in the cross-state regression of state NFCS literacy on mandate exposure (largest candidate $F = 7.06$, wrong direction). A cohort-matched implementation using NFCS micro-data also fails: with state, wave, and band fixed effects, the cohort-specific first-stage $F = 0.05$. The mandate variable is endogenous to state literacy through a reverse-causality channel (low-literacy states adopted mandates as a remedy). The OLS estimate is the

preferred inferential statement. A clean IV strategy would require pre-1990 state economics-education mandate variation [Bernheim et al., 2001] or NFCS micro-data restricted to cohorts demonstrably treated under a verifiable-compliance mandate; neither is straightforward.

Ownership-interactive state-level confounds. The within-stock contrast does not discriminate against a state-level moderator that itself operates differently across IO terciles. The canonical instance is state-resident public-pension-fund home-state ownership, where state pensions concentrate holdings in mid-IO firms with mechanical state-level co-movement. A keyword filter of the 13F manager universe identifies ≈ 21 distinct state-pension-fund managers concentrated in roughly seven states with active state-pension systems; for the remaining ≈ 44 states, home-state-pension residence is collinear with the state fixed effect. State-employee 401(k) custodians and state-resident-brokerage concentration are the closest extensions of this confound class. Closing the class requires a CIK-by-CIK state-pension-resident flag matched against the 13F panel; that data source is not built here.

Functional-form fragility of the U-shape. The continuous-margin quadratic $\hat{\gamma}_5 = +0.0151$ ($p = 0.017$) does not survive log-IO ($p = 0.75$), kink-IO ($p = 0.24$), or a flat twenty-bin diagnostic. The trough-location estimate $IO^* \approx 0.34$ is correspondingly parametric. The discrete-tercile mid-IO concentration is the headline functional-form-free characterization; the U-shape is its parametric corroboration only.

8 Conclusion

A large literature documents that less financially literate investors trade systematically worse, and a separate large literature documents that retail demand is geographically concentrated and moves prices. Put those two facts together and the prediction is direct: state-level investor literacy should moderate the return anomalies that depend on retail mispricing. Aggregate tests on U.S. equity panels deliver weak signals. The contribution of this paper is to show that the literacy mechanism is not absent — it is hidden by aggregation. Where in the cross-section the mechanism operates, and under what conditions on ownership and size, is the question this paper sets out to answer.

The data identify an interaction zone. The literacy channel requires two ingredients to leave a price footprint: retail demand large enough to move prices, and institutional activity sufficient to translate that demand into return differences detectable across literate and illiterate states. Where neither ingredient is present (micro-cap firms held mostly by retail), mispricing accumulates but does not load on state-literacy variation. Where institutions dominate (large firms with high IO), arbitrage absorbs retail demand and neutralizes the local-state channel. The interaction zone — firms with retail demand at meaningful market size and enough institutional activity to arbitrage at the margin — is where the literacy channel can be detected, and is where the data find the moderation. In the full stable-HQ panel, the interac-

tion zone falls predominantly in mid-IO. Restricting the panel to larger firms relocates it to low-IO, and the cross-state literacy gradient migrates with it. Firms outside the interaction zone show no gradient at all.

The empirical signature of this story — migration of the moderation with sample composition — is the central new piece of evidence the paper introduces. The IO discretization is sample-conditional; the underlying economic population is not. The migration evidence identifies the interaction-zone reading without requiring an instrumental-variables strategy, which is fortunate because the natural candidate instrument (state K-12 personal-finance mandates) fails its first-stage relevance test for documented reasons: low-literacy states adopted mandates in response to low adult literacy, and a cohort-matched implementation also fails. The headline result rests on OLS identification under the conditional-unconfoundedness assumption, and the migration test provides the predicted-signature confirmation that an instrument would otherwise be asked to provide.

Three honest limitations bound the result. The continuous-margin U-shape clears five percent statistical significance on a quadratic specification only; log-IO, kink-IO, and a nonparametric twenty-bin diagnostic all return null curvature. The functional-form-free claim is the discrete-tercile contrast, and the quadratic is corroboration only. The point-in-time literacy reassignment on the full panel collapses the headline coefficient by roughly twenty-six fold, leaving the stable-headquarters subsample as the population for which the characterization holds; this is the binding population-scope

limitation. The within-mid-IO IV-decile decomposition fails to reject decile equality, rejecting the textbook reading that arbitrage costs binding at high IV should generate the strongest moderation at the highest IV decile. The microfoundation the paper supports is identified through size and ownership conditioning, not through the IV margin.

The result has implications for how we read state-level investor characteristics in asset pricing. Whether other state-level investor traits — household participation rates, brokerage density, retail-attention measures — operate through a similar interaction zone is a natural next question, and the size-restriction migration test developed here provides a portable empirical signature for testing it. Whether the within-family microfoundation is best read as a demand-intensity channel, an attention channel, or a clientele-timing channel is a question the present design treats as observationally similar. The TAQ-based Boehmer-Jones-Zhang retail trade indicator would discriminate among these readings; the data construction for that test is a natural next-paper extension. Pre/post-relocation event studies on the 1,366 firms dropped from the stable-HQ sample would discriminate between the mechanism-strength and sample-selection readings of the PIT magnitude collapse. The interaction zone, characterized as it is, suggests where to look next.

A Variable construction detail

Extended robustness material is in the Internet Appendix: the Herfindahl-Hirschman alternative (IA ??), the Korniotis-Kumar business-cycle block (IA ??), the difference-in-differences secondary design (IA ??), the full PIT literacy reassignment (IA ??), the relocater-vs-stable-HQ characteristics comparison (IA ??), the Thomson Reuters s34 panel construction (IA ??), and alternative literacy constructions (IA ??).

NFCS Big Five share-correct score. The state financial-literacy measure is the share of NFCS respondents in a state-wave who answer the FINRA Big Five literacy questions correctly. Correct-answer keys from the FINRA codebook: M6=1, M7=3, M8=2, M9=2, M10=1. An earlier build coded all five questions with a single key; the corrected keys are used throughout, and the Table 1 state-literacy rankings are post-correction. State-wave scores are forward-filled to monthly frequency between waves, then z-scored within calendar month to form `literacy_z`.

Idiosyncratic volatility. Standard deviation of daily Fama-French three-factor residuals over a rolling twelve-month window ending in the observation month; z-scored within calendar month before entering the interaction terms.

References

- B. Douglas Bernheim, Daniel M. Garrett, and Dean M. Maki. Education and saving: The long-term effects of high school financial curriculum mandates. *Journal of Public Economics*, 80(3):435–465, 2001. doi: 10.1016/S0047-2727(00)00120-1.
- Gennaro Bernile, Alok Kumar, and Johan Sulaeman. Home away from home: Geography of information and local investors. *Review of Financial Studies*, 28(8):2009–2049, 2015. doi: 10.1093/rfs/hhv004.
- Laurent E. Calvet, John Y. Campbell, and Paolo Sodini. Down or out: Assessing the welfare costs of household investment mistakes. *Journal of Political Economy*, 115(5):707–747, 2007. doi: 10.1086/524204.
- A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427, 2008. doi: 10.1162/rest.90.3.414.
- Joshua D. Coval and Tobias J. Moskowitz. Home bias at home: Local equity preference in domestic portfolios. *Journal of Finance*, 54(6):2045–2073, 1999. doi: 10.1111/0022-1082.00181.
- Mark Grinblatt, Matti Keloharju, and Juhani Linnainmaa. Iq, trading behavior, and performance. *Journal of Financial Economics*, 104(2):339–362, 2012. doi: 10.1016/j.jfineco.2011.05.016.

Alexander Hillert, Heiko Jacobs, and Sebastian Müller. Media makes momentum. *Review of Financial Studies*, 27(12):3467–3501, 2014. doi: 10.1093/rfs/hhu061.

Harrison Hong, Jeffrey D. Kubik, and Jeremy C. Stein. The only game in town: Stock-price consequences of local bias. *Journal of Financial Economics*, 90(1):20–37, 2008. doi: 10.1016/j.jfineco.2007.11.006.

Zoran Ivković and Scott Weisbenner. Local does as local is: Information content of the geography of individual investors’ common stock investments. *Journal of Finance*, 60(1):267–306, 2005. doi: 10.1111/j.1540-6261.2005.00730.x.

Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1):65–91, 1993. doi: 10.1111/j.1540-6261.1993.tb04702.x.

George M. Korniotis and Alok Kumar. State-level business cycles and local return predictability. *Journal of Finance*, 68(3):1037–1096, 2013. doi: 10.1111/jofi.12017.

George M. Korniotis, Alok Kumar, and Jeremy K. Page. Investor sophistication and asset prices. *Review of Financial Economics*, 38(4):557–579, 2020. doi: 10.1002/rfe.1093.

Annamaria Lusardi and Olivia S. Mitchell. The economic importance of

- financial literacy: Theory and evidence. *Journal of Economic Literature*, 52(1):5–44, 2014. doi: 10.1257/jel.52.1.5.
- Christo Pirinsky and Qinghai Wang. Does corporate headquarters location matter for stock returns? *Journal of Finance*, 61(4):1991–2015, 2006. doi: 10.1111/j.1540-6261.2006.00895.x.
- Jeffrey Pontiff. Costly arbitrage and the myth of idiosyncratic risk. *Journal of Accounting and Economics*, 42(1–2):35–52, 2006. doi: 10.1016/j.jacceco.2006.04.002.
- Andrei Shleifer and Robert W. Vishny. The limits of arbitrage. *Journal of Finance*, 52(1):35–55, 1997. doi: 10.1111/j.1540-6261.1997.tb03807.x.
- Carly Urban. Specialized courses in personal finance for high school students. *Journal of Financial Counseling and Planning*, 31(2):236–251, 2020. doi: 10.1891/JFCP-19-00059.
- Matthew D. Webb. Reworking wild bootstrap based inference for clustered errors. *Queen’s Economics Department Working Paper*, (1315), 2014.

Table 1: Descriptive statistics and state literacy rankings
Panel A: Distribution of focal variables (full panel)

Variable	Mean	SD	p05	p50	p95
Monthly return	0.013	0.192	−0.220	0.005	0.253
Momentum (<code>mom_12_2</code>)	0.107	0.828	−0.678	0.034	1.021
Idiosyncratic volatility	0.027	0.022	0.007	0.020	0.070
Market equity (\$ billions)	7.04	42.1	0.017	0.596	26.19
NFCS Big Five share correct	0.244	0.028	0.200	0.247	0.286

Panel B: Panels and merge conventions

Panel	Firm-months	Firms
Full panel (static-snapshot HQ, baseline literacy merge)	623,896	7,447
Full panel, PIT literacy reassignment	623,814	7,447
Stable-HQ subsample (no in-sample EDGAR relocation)	519,090	6,081
with persistent IO measure	518,962	6,018
with time-varying IO measure	493,889	5,877

Panel C: State literacy rankings and NFCS wave summary

Highest-literacy states	MN, HI, NH, DC, CT
Lowest-literacy states	MS, OH, AR, TX, WV
NFCS wave means (2009, 2012, 2015, 2018, 2021)	0.258, 0.254, 0.255, 0.229, 0.222
Median within-state rank correlation across waves	0.71 (Spearman)

Notes. Idiosyncratic volatility is the standard deviation of daily Fama-French three-factor residuals over a rolling twelve-month window. Market equity is price times shares outstanding from CRSP, in billions of dollars. The NFCS Big Five share-correct score is the state-wave share of respondents answering the FINRA Big Five financial-literacy questions correctly. The within-state rank correlation across the five NFCS waves is the median Spearman correlation of state-wave ranks across consecutive waves (a measurement-stability statistic for the state literacy ordering).

Table 2: Mid-IO concentration: three-way coefficient by IO tercile, stable-HQ subsample

<i>Panel A: Persistent IO measure</i>					
Tercile	Mean IO share	$\hat{\gamma}$	State-clustered t	wcb p	N
IO1 (low; high retail)	0.170	−0.0113	−1.04	0.296	144,989
IO2 (mid)	0.495	−0.0321	−2.63	0.064	191,589
IO3 (high; institutional)	0.704	−0.0050	−0.31	0.760	182,384
<i>Panel B: Time-varying IO measure</i>					
Tercile	Mean IO share	$\hat{\gamma}$	State-clustered t	wcb p	N
IO1 (low; high retail)	0.174	−0.0087	−0.92	0.337	141,129
IO2 (mid)	0.504	−0.0265	−2.28	0.058	184,881
IO3 (high; institutional)	0.711	−0.0044	−0.25	0.789	167,879

Notes. Each row reports the three-way coefficient $\hat{\gamma}$ on $\text{mom} \times \text{IV} \times \text{literacy_z}$ estimated within the IO tercile, on the stable-HQ subsample under static-snapshot literacy. The persistent measure is each firm’s time-mean institutional-ownership share; the time-varying measure assigns each firm-month the share from the most recent 13F quarter with report date at least four months earlier. Wild-cluster bootstrap uses Webb six-point weights [Cameron et al., 2008] with $B = 4,999$ and null imposed. Sample: January 2009 through December 2023.

Table 3: Size-controlled mid-IO three-way: literacy survives

IO tercile	Baseline $\hat{\gamma}$ (state- t)	Size-controlled $\hat{\gamma}$ (state- t)	N
IO1 (low)	−0.0113 (−1.03)	−0.0077 (−0.65)	144,989
IO2 (mid)	−0.0321 (−2.61)	−0.0256 (−2.24)	191,589
IO3 (high)	−0.0050 (−0.31)	−0.0001 (−0.00)	182,384
Horse race in mid-IO: γ_{size} on $\text{mom} \cdot \text{IV} \cdot \text{size_z}$	—	+0.0017 (+0.64)	191,589

Notes. Baseline column re-estimates Equation (1) within IO tercile. Size-controlled column estimates Equation (2) within IO tercile, adding size_z , $\text{mom} \cdot \text{size_z}$, $\text{IV} \cdot \text{size_z}$, and $\text{mom} \cdot \text{IV} \cdot \text{size_z}$. State and month fixed effects throughout. State-clustered t in parentheses. The horse-race row reports the coefficient on size^3 three-way under the size-controlled specification within mid-IO.

Table 4: Interaction-zone migration: IO tercile three-way coefficient under size filters, size-controlled spec

Panel A: Within-panel three-way coefficient $\hat{\gamma}_{lit}$ by IO tercile

Size filter	IO1 (low)	IO2 (mid)	IO3 (high)	N
Full stable-HQ panel (paper headline)	-0.0077 (-0.65)	-0.0256 (-2.24)	-0.0001 (-0.00)	518,962
Drop bottom 20% (micro-cap)	-0.0137 (-1.14)	-0.0402 (-2.57)	+0.0109 (+0.75)	454,107
Drop bottom 33% (small-cap)	-0.0155 (-1.02)	-0.0273 (-1.29)	+0.0122 (+0.66)	395,873
Drop bottom 50% (below median)	-0.0441 (-2.04)	-0.0098 (-0.58)	+0.0115 (+0.58)	322,137

Panel B: Cross-state weighted slope of per-state $\hat{\gamma}_{lit}$ on state literacy

Sample cell	Weighted slope	state- t	Pearson r	n_{states}
Full panel, mid-IO (headline cell)	-0.0623	-5.01	-0.275	42
Drop bottom 50%, mid-IO	-0.0325	-3.82	+0.008	31
Drop bottom 50%, IO1 (migrated headline cell)	-0.0379	-4.30	-0.291	34
Drop bottom 50%, IO3 (predicted null)	+0.0061	+0.26	+0.054	31

Notes. Panel A: Equation (2) within each IO tercile, under each size filter applied at the firm level using firm panel-mean market capitalization. State and month fixed effects throughout. State-clustered t in parentheses. Bold cells indicate the tercile carrying the literacy moderation under each filter. Panel B: per-state $\hat{\gamma}_{lit}$ from Equation (2) with month fixed effects only (state fixed effects collapse for single-state regressions); the cross-state regression of $\hat{\gamma}_{lit}$ on state NFCS literacy uses precision weighting ($1/\text{SE}^2$). States with fewer than 5 firms in the sample cell are excluded. The cross-state state- t reported is the precision-weighted-slope t -statistic.

Table 5: Within-mid-IO IV-decile decomposition (persistent IO, stable-HQ, size-controlled)

IV decile	$\hat{\gamma}$ (no size ctrl)	$\hat{\gamma}$ (size-controlled)	N
1 (lowest IV)	+0.026	+0.068	28,448
2	−0.003	+0.003	23,732
3	+0.266 ($t = +2.20$)	+0.267 ($t = +2.20$)	21,634
4	−0.119	−0.101	20,435
5	+0.010	+0.019	19,200
6	−0.091	−0.078	18,737
7	−0.279 ($t = -2.79$)	−0.279 ($t = -2.88$)	17,684
8	−0.119	−0.113	16,315
9	−0.009	+0.011	14,113
10 (highest IV)	−0.045	−0.035	11,291
Joint $\chi^2(9)$ equality test	$p = 0.94$ (paper)	—	—
Spearman $\rho(\text{decile}, \hat{\gamma})$	—	−0.67 ($p = 0.033$)	—

Notes. Each row is $\hat{\gamma}$ on `mom × IV × literacy_z` within mid-IO restricted to the within-month NYSE IV decile. State and month fixed effects throughout; permno-clustered t . The size-controlled column adds the four size-moderation controls. The joint chi-square equality test is the original Table 2-spec test; the Spearman correlation summarizes the broader within-decile pattern under the size-controlled spec.

Table 6: The nested baseline: aggregate and mid-IO three-way under PIT correction

<i>Panel A: Aggregate three-way coefficient</i>				
Specification	$\hat{\gamma}$	State- t	wcb p	N
Full panel, static-snapshot literacy	−0.0117	−2.17	0.078	623,896
Full panel, PIT literacy reassignment	−0.0003	−0.86	0.517	623,814
Stable-HQ subsample, static literacy	−0.0145	−1.94	0.122	519,090

<i>Panel B: Mid-IO coefficient under PIT, with and without size controls</i>				
Specification	$\hat{\gamma}_{\text{mid-IO}}$	State- t	Collapse vs. baseline	N
Stable-HQ static, persistent (paper headline)	−0.0321	−2.63	—	191,589
Stable-HQ static, size-controlled	−0.0256	−2.24	—	191,589
PIT full panel, persistent	−0.00121	−2.07	$\approx 27\times$ smaller	191,589
PIT full panel, size-controlled	−0.00100	−1.75	$\approx 26\times$ smaller	191,589

Notes. Static-snapshot merge pairs each firm-month with the literacy of its Compustat **state**; PIT reassignment re-merges against the firm’s then-current 10-K-header state and re- z -scores within month. State-level wild-cluster bootstrap (Webb six-point, $B = 4,999$ or $9,999$, null imposed). The PIT magnitude collapse is essentially the same with and without size controls ($27\times$ vs. $26\times$).

Table 7: Functional-form grid: the quadratic five-percent clearance is parametric

Specification	Focal term	Coef	State- t	wcb p
Quadratic (corroborating)	γ on IO^2	+0.0151	+2.28	0.017
Cubic	γ on IO^3	+0.798	+1.70	0.133
Log-IO	γ on $\log(\text{IO})$	−0.00243	−0.35	0.748
Kink-IO at 0.30	kink coef	+0.179	+1.19	0.242
20-bin diagnostic	conditional means within ± 0.003 of zero	—	—	—

Notes. Stable-HQ persistent-IO subsample. Each row reports the focal-basis coefficient with state-clustered t and wild-cluster bootstrap p . The 20-bin diagnostic partitions persistent IO into 20 quantile bins, residualizes the focal triple’s contribution to monthly returns on state and month fixed effects within each bin, and reports the within-bin conditional mean of the demeaned composite.

Table 8: Cross-state literacy gradient by sub-period (size-controlled, mid-IO)

Period	Weighted slope	state- t	Pearson r	n_{states}
Pre-2015 (2009–2014)	+0.0403	+1.89	+0.048	37
Post-2015 (2015–2023)	−0.0706	−3.50	−0.370	38
Pooled (2009–2023)	−0.0623	−5.01	−0.275	42
NFCS Wave 2009 only (2009–2011)	+0.281	+2.57	+0.109	34
NFCS Waves 2012–2021	−0.114	−1.18	−0.066	—

Notes. Per-state $\hat{\gamma}_{\text{lit}}$ from Equation (2) with month fixed effects only; precision-weighted cross-state regression. States with ≥ 5 firms in the sample cell. Wave-2009 row segregates the first NFCS administration (which produced the wrong-signed cross-state gradient driving the entire pre-2015 result).

Table 9: First-stage diagnostics: NFCS state literacy on mandate exposure

Instrument	Slope	Pearson r	First-stage F
has_pf_mandate (ever-treated by 2024)	−1.31	−0.355	7.06
early_pf_adopter (treated by 2000)	+0.72	+0.112	0.62
cohort_share_affected	−0.59	−0.051	0.13
log_years_pf	−0.27	−0.171	1.47

Notes. Cross-state regression of state NFCS literacy (panel mean %) on each candidate instrument, $n = 51$ states. The strongest candidate (has_pf_mandate) has $r = -0.355$ (wrong sign for a positive IV) and $F = 7.06$ (below the Stock-Yogo $F > 10$ threshold).

The mandate-literacy correlation is negative because the South and Appalachia (low-literacy regions) adopted mandates as a remedy; the Northeast and Pacific (high-literacy regions) did not. This is reverse causality, not a flaw in the underlying literacy story.